

Title: Rod photoreceptors but not cones demonstrate the ability to recover from acute hypoxia in the mouse retina.

Purpose: It is generally accepted in the field that photoreceptors are highly energetically demanding and rely on anaerobic glycolysis for ATP production. Cones have 3 times more mitochondria than rods and greater metabolic demand. Despite these higher energy requirements, it is unknown whether cones and rods differ in their vulnerability to hypoxia. We address this gap in knowledge by examining ex vivo ERG light responses under acute hypoxia in the mouse retina.

Methods: We addressed this question by measuring light responses using ex vivo electroretinogram (ERG) after 3 hours of hypoxia as well as after a 12-hour reoxygenation period. To isolate rod and cone responses, experiments were done using retinas from *GNAT1*^{-/-} and *GNAT2*^{-/-} mice. *GNAT1*^{-/-} mice lack functional rod photoreceptors while *GNAT2*^{-/-} mice lack functional cone photoreceptors. We used pharmacological blockers BaCl₂ (100 μM) to block the slow PIII component of Müller glia which masks photoreceptors responses and DL-AP4 (40 μM) to isolate ON-bipolar cell responses.

Results: Mixed effects analysis revealed that normalized A wave amplitudes significantly decreased in both genotypes after 3 hours of hypoxia. However, *GNAT1*^{-/-} retinas retained significantly higher normalized amplitudes compared to *GNAT2*^{-/-} retinas in both 3 hours of hypoxia (p = 0.021, N = 5) and hypoxia + reoxygenation conditions (p = 0.0007, N = 5) indicating initial resistance. However, only rod photoreceptors recovered some amplitude after 12 hours of reoxygenation. We examined inner retina function in relation to photoreceptor function by looking at A: B wave ratios. A:B wave ratios remained unchanged in *GNAT1*^{-/-} retinas indicating that cone ON-Bipolar cell loss is proportional to cone photoreceptor loss. In contrast, the A:B wave ratio significantly increased in *GNAT2*^{-/-} retinas after 3 hour of hypoxia (p = 0.0234, N = 5) indicating that rod bipolar cell loss cannot be fully explained by loss of rod responses. To examine how synaptic signaling is impacted by hypoxia we show that there is a 40% decrease in docked synaptic vesicles at the ribbon synapse in both cell types as well as altered mitochondria structure.

Conclusions: While both rod and cone photoreceptors are severely impacted by 3 hours of hypoxia only rods show some recovery after 12 hours of reoxygenation, while cone photoreceptors initially lose light responses to a lesser degree than rod photoreceptors. In addition, A:B wave ratio reveals differences in cone and rod pathway inner retina vulnerabilities to hypoxia as rod bipolar cell loss cannot be fully explained by rod photoreceptor amplitude loss.